# Benjamin Ponce-Martinez

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## EDUCATION

**Iowa State University** Ames, IA May 2027
*Bachelor of Science, Electrical Engineering*

## WORK EXPERIENCE

**IINSPIRE LSAMP, Iowa State University Department of Mechanical Engineering** Aug 2022 – Aug 2023
*Undergraduate Research Assistant*

- Designed and fabricated acoustic devices integrating ultrasonic fields into additive manufacturing platforms for advanced 3D printing applications
- Developed and optimized an acoustic flow cell capable of extruding internally structured composite materials with improved energy efficiency
- Modeled and fabricated millimeter-scale fluid channels using 3D CAD design and stereolithography printing techniques for fluid mechanics research

**IINSPIRE LSAMP VSEER, Iowa State University Department of Mechanical Engineering** May 2022 – Aug 2022
*Undergraduate Research Assistant*

- Designed and operated glass-based capillary micromodels to study oil displacement and porous media flow behavior under microscope imaging
- Improved micromodel fabrication processes to enhance durability and experimental reliability
- Prepared custom fluid solutions and performed controlled flow visualization experiments to optimize imaging clarity and data accuracy

## PROJECTS

Eddy Current Probes for Electrical Conductivity Measurements – Senior Design Aug 2026 – May 2027

- Design and optimize eddy current probes to measure the electrical conductivity of metallic materials using a low-cost Vector Network Analyzer (VNA)
- Analyze S-parameters and coil impedance variations to evaluate conductivity and magnetic permeability
- Calibrate and validate the measurement system using reference samples with known conductivity values to improve measurement accuracy and reliability

Semiconductor Fabrication and Characterization – EE 4320 Aug 2026 – Dec 2026

- Fabricated semiconductor device structures on silicon wafers using thermal oxidation, photolithography, wet etching, and boron/phosphorus diffusion processes
- Operated semiconductor fabrication equipment, including oxidation furnaces, spin coaters, and mask aligners, to pattern NMOS and PMOS transistor regions
- Performed wafer characterization using a Filmetrics reflectometer and four-point probe, measuring oxide thickness and sheet resistance while calculating resistivity and doping concentrations.

Temperature Indicator Circuit – EE230 Design Lab Aug 2024 – Dec 2024

- Designed and built an analog temperature indicator circuit using an LM35 sensor and op-amp comparators to control multi-color LED outputs
- Implemented hysteresis to ensure stable LED transitions and accurate response to rising and falling temperatures
- Calibrated and validated circuit thresholds at 30°C, 40°C, 80°C, and 100°C using a variable DC voltage source

## SKILLS

- Technical Skills: circuit designing, microsoft office, solidworks, python, java, C, verilog
- Soft Skills: leadership, time management, team player, attention to detail